Piyush Purohit

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SUMMARY

- Software Engineer with 4+ years building and shipping production ML systems, including low-latency model serving, automated LLM evaluation and hallucination guardrails, graph-based abuse detection, and semantic retrieval. At Meta, build Trust & Safety enforcement and evaluation infrastructure that measures over-enforcement and gates ML experimentation. Previously built real-time embedding-based ad matching and Kafka/Flink feature pipelines at Media.net. Strong in Python, Java, Go, and distributed systems on Kubernetes and AWS.

WORK EXPERIENCE

Software Engineer **Menlo Park, CA**
Meta *04/2025 to 10/2026*

- Designed an automated LLM evaluation and labeling framework with a four-stage anti-hallucination verification pipeline, establishing the team's first system for systematically measuring over-enforcement and enabling evaluation-driven ML experimentation.
- Engineered a low-latency, multi-task ML model-serving pipeline for real-time scoring of negative user feedback, driving a 2% lift in engagement and an 8% boost in integrity while cutting six-figure annual infrastructure costs.
- Architected a unified, multi-surface Trust & Safety enforcement framework powering Feed, Search, and Notifications, reducing new harm-type deployment from multiple independent rollouts to a single parameterized rule set.
- Migrated legacy, bespoke enforcement callsites onto the shared framework using shadow-mode traffic diffing, regression testing, and parity validation, eliminating divergent policy logic with zero downtime.
- Architected a multi-tier recidivist detection engine using device-linked graph analysis and tiered confidence modeling to identify and neutralize coordinated inauthentic behavior at the account level.

Software Engineer II **Mumbai, India**
Media.net *01/2021 to 08/2023*

- Architected a real-time keyword ad-serving engine using Universal Sentence Encoder embeddings and Redis, enabling sub-20 ms bidding decisions and increasing ad revenue by 150%.
- Built a real-time distributed data ingestion and feature pipeline with Apache Flink, Kafka, and Aerospike as a low-latency lookup store, processing 200K+ event records per day for downstream ML consumers.
- Engineered high-throughput Reporting APIs in Java Spring Boot with customized thread pools to aggregate ad metrics across multi-node OLAP clusters for 1M+ records, reducing latency by 25%.
- Containerized microservices with Docker and Kubernetes and established Jenkins CI/CD pipelines for automated testing and zero-downtime rollouts; managed AWS infrastructure with Terraform, auto-scaling, and Kibana/Logstash observability.

Software Engineer Intern **San Ramon, CA**
Career Launch *01/2025 to 03/2025*

- Architected and deployed core backend services, database schemas, and AWS infrastructure from scratch, scaling to 10K+ users and 100K+ daily records.

Research Assistant **Santa Clara, CA**
Santa Clara University *01/2024 to 06/2024*

- Built a compiler engine for parsing recursive queries in the distributed database system Apache AsterixDB.

TECHNICAL SKILLS

- **Languages**: Python, Java, Go, C++, SQL, TypeScript, Rust, FastAPI, Spring Boot, Django, gRPC, Gin, REST APIs, microservices.
- **Machine learning**: TensorFlow, ResNet18, Universal Sentence Encoder, multi-task learning, model serving and monitoring, real-time scoring, deep learning, computer vision, ranking, recommendation systems, optimization for cost and latency.
- **LLM systems**: prompt engineering, tool calling, RAG, semantic search, embeddings, vector search, approximate nearest neighbors (ANN), guardrails, hallucination detection, LLMOps, MLOps.
- **Agents**: AI agents, agentic workflows, LangGraph, LangChain, tool calling, workflow orchestration.
- **Evaluation and quality**: LLM evaluation, automated evals, regression testing, shadow-mode traffic diffing, parity validation, model-quality measurement, online experimentation, production monitoring.
- **Infrastructure**: AWS (EC2, Lambda, S3, RDS), Docker, Kubernetes, Terraform, Jenkins, CI/CD, Kafka, Flink, Spark, Redis, Aerospike, RabbitMQ, PostgreSQL, MongoDB, Qdrant, Kibana, Logstash, stream processing, feature pipelines, observability. AWS Solutions Architect Associate (2025).

EDUCATION

Santa Clara University **Santa Clara, CA**
Master of Science in Computer Science and Engineering *09/2023 to 06/2025*

University of Mumbai **Mumbai, India**
Bachelor of Engineering in Computer Science *08/2016 to 11/2020*

PROJECTS

- **Visual product recommendation system**. Built a retrieval pipeline using ResNet18 image embeddings and approximate nearest-neighbor search over the Kaggle H&M product dataset, with Cypher as the vector store, to power similarity-based recommendations.
- **MiniMQ, distributed message broker**. Designed a fault-tolerant broker in Java with leader election, heartbeat, and gossip protocols, persistent storage, and dynamic load balancing; deployed on four AWS EC2 instances at 1,600 QPS.